DATA SCIENCE

MODULE 3

FEATURE SELECTION AND DECISION TREE FOUNDATIONS

Study Notes • Definitions • Algorithms • Worked Examples • Exam Questions

Learning objectives

- Understand why feature selection is important in machine learning.
- Compare filter, wrapper, and embedded feature-selection methods.
- Explain entropy, information gain, and Gini impurity.
- Understand how a decision tree chooses splits and makes predictions.
- Describe overfitting, pruning, and practical evaluation.

Prepared as beginner-friendly classroom and exam revision notes.

1. Feature Selection

Definition: Feature selection is the process of choosing a useful subset of the available input variables (features) for building a machine-learning model. It removes irrelevant, redundant, or noisy features while retaining information useful for prediction.

Why is feature selection needed?

• **Improves generalization:** removing noise can help a model perform better on unseen data.

• **Reduces overfitting:** fewer unnecessary inputs can reduce the chance of learning accidental patterns.

• **Faster training:** fewer features can reduce computation and memory requirements.

• **Improves interpretability:** a smaller set of features is easier to explain.

• **Reduces data-collection cost:** in some applications, fewer measurements need to be collected.

Feature selection vs. feature extraction

Feature Selection	Feature Extraction
Selects some of the original features.	Creates new features by transforming or combining existing ones.
Original meaning is usually preserved.	New features may be harder to interpret.
Example: choose Age and Income from 20 columns.	Example: PCA transforms many variables into principal components.

Types of feature-selection methods

A. Filter methods

Filter methods score features using statistical properties of the data, without training a specific predictive model for every candidate subset. They are usually fast and model-independent.

• **Variance threshold:** removes features with almost no variation.

• **Correlation:** identifies features strongly related to the target; correlation between input features can also reveal redundancy.

• **Chi-square test:** commonly used to test association between categorical features and a categorical target; inputs must satisfy the method's requirements.

• **ANOVA F-test:** can compare the means of a numeric feature across target classes.

• **Mutual information:** measures how much knowing a feature reduces uncertainty about the target; it can detect nonlinear relationships.

Advantages: fast, simple, scalable. **Limitations:** may ignore interactions between features and the chosen model.

B. Wrapper methods

Wrapper methods evaluate feature subsets by training and validating a predictive model. They can capture how features work together, but may be computationally expensive.

• **Forward selection:** start with no features; add the feature that improves validation performance the most, repeating until a stopping rule is met.

• **Backward elimination:** start with all candidate features; remove the least useful feature step by step.

• **Recursive Feature Elimination (RFE):** fit a model, rank features by importance, remove the least important ones, and repeat.

Advantages: considers the model and feature combinations. **Limitations:** slower and more prone to overfitting if validation is not done carefully.

C. Embedded methods

Embedded methods perform feature selection as part of model training.

• **Lasso (L1 regularization):** can shrink some linear-model coefficients to exactly zero.

• **Decision trees:** select features for splits based on impurity reduction.

• **Random forests:** provide feature-importance measures aggregated across many trees; importance should be interpreted carefully, especially with correlated features.

Quick comparison: Filter = score features before model training; Wrapper = search subsets using model performance; Embedded = selection occurs during model training.

Important precautions

• Split the data into training and test sets before feature selection.

• Fit selection rules only on the training data, ideally inside each cross-validation fold, to avoid data leakage.

• Do not select features by looking at test-set performance.

• A feature's usefulness depends on the target, data quality, and model; importance does not automatically mean causation.

2. Decision Tree Foundations

Definition: A decision tree is a supervised-learning algorithm that makes predictions by repeatedly splitting data according to feature-based rules. It can be used for classification (predicting categories) and regression (predicting numeric values).

Main parts of a decision tree

• **Root node:** the first node, representing the full training dataset.

• **Internal/decision node:** a test on a feature, such as “Age ≤ 30?”.

• **Branch:** an outcome of a test, such as Yes/No.

• **Leaf/terminal node:** the final prediction, such as “Approved” or a numeric value.

How a decision tree works

1. Begin with the training examples at the root.
2. Evaluate possible splits on available features.
3. Choose a split that makes the child groups more pure or reduces prediction error.
4. Repeat the process within each child node.
5. Stop when a stopping condition is met; assign a prediction to each leaf.

Example rule path: If **Income = High** and **Credit History = Good**, then predict **Loan = Approve**. The tree is a collection of such decision rules.

3. Entropy and Information Gain

Entropy

Entropy measures the impurity or uncertainty of class labels in a node. For a classification node with class probabilities p■, p■, …, p■:

Entropy(S) = – Σ p■ log■(p■)

For binary classification, entropy is 0 when every example belongs to one class and reaches 1 when the two classes are equally represented. A common convention is to treat 0 × log■(0) as 0.

Worked entropy example

Suppose a node contains 10 students: 6 Pass and 4 Fail. Then p(Pass)=0.6 and p(Fail)=0.4.

Entropy = –[0.6 log■(0.6) + 0.4 log■(0.4)] ≈ 0.971 bits.

If all 10 students passed, entropy would be 0 because the node contains only one class.

Information Gain

Information Gain (IG) measures how much a split reduces entropy. It is the parent entropy minus the weighted average entropy of the child nodes.

IG(S, A) = Entropy(S) – Σ (|S■| / |S|) Entropy(S■)

The split with the highest information gain is preferred by the basic ID3 decision-tree algorithm.

Worked information-gain example

A dataset has 10 examples (6 Pass, 4 Fail), so parent entropy is approximately 0.971. A candidate split creates:

• Left child: 5 examples (all Pass), entropy = 0.

• Right child: 5 examples (1 Pass, 4 Fail), entropy ≈ 0.722.

Weighted child entropy = (5/10 × 0) + (5/10 × 0.722) = 0.361. Information Gain ≈ 0.971 – 0.361 = **0.610 bits**.

4. Gini Impurity and Split Criteria

Gini impurity is another measure of class mixing, commonly used by CART (Classification and Regression Trees).

Gini(S) = 1 – Σ p■²

For the 6 Pass / 4 Fail node: Gini = 1 – (0.6² + 0.4²) = 0.48. Gini is 0 for a pure node. A split is evaluated by the weighted impurity of its children; lower weighted Gini is better.

Criterion	Meaning	Typical use
Information Gain	Reduction in entropy after a split.	ID3
Gain Ratio	Information gain adjusted for the split's intrinsic information.	C4.5
Gini Impurity	Probability-based measure of class mixing.	CART classification trees
Variance / squared error reduction	Reduction in variation of numeric target values.	Regression trees

5. Common Decision-Tree Algorithms

ID3

ID3 builds a classification tree by choosing the feature with the highest information gain at each step. The original form is mainly associated with categorical attributes and can favor features with many distinct values.

C4.5

C4.5 extends ID3. It uses gain ratio, can handle continuous attributes by choosing thresholds, and includes pruning techniques.

CART

CART stands for Classification and Regression Trees. It typically creates binary splits, uses Gini impurity for classification and squared-error-based criteria for regression, and supports pruning.

Algorithm	Split criterion / characteristic
ID3	Information gain; classic version mainly categorical inputs.
C4.5	Gain ratio; supports continuous attributes and pruning.
CART	Binary splits; Gini for classification and error reduction for regression.

6. Overfitting, Stopping Rules, and Pruning

A very deep tree may memorize noise in the training data. It can achieve excellent training accuracy but perform poorly on new data; this is called overfitting.

Ways to control overfitting

• **Maximum depth:** limits how many levels the tree can grow.

• **Minimum samples per split/leaf:** prevents decisions based on very small groups.

• **Minimum impurity decrease:** requires a split to provide enough improvement.

• **Pre-pruning:** stop growth early using constraints.

• **Post-pruning:** grow a larger tree and then remove branches that do not improve generalization.

Underfitting occurs when a tree is too simple to capture useful patterns. Choose complexity using validation or cross-validation, then evaluate the final model once on held-out test data.

7. Advantages and Limitations of Decision Trees

Advantages	Limitations
Easy to visualize and explain.	Deep trees can overfit.
Can model nonlinear relationships and feature interactions.	Small data changes can produce a different tree.
Little feature scaling is usually needed.	Greedy split choices may not produce a globally optimal tree.
Can be used for classification and regression.	Some split criteria can be biased toward features with many possible values.

8. Feature Selection and Decision Trees Together

Decision trees can help with feature selection because they choose features that improve splits. Features never used in a fitted tree may be less useful for that particular tree, but this does not prove they are universally irrelevant. Correlated features can substitute for one another, and importance rankings may change across datasets.

• Use a training/validation process to compare feature subsets.

• Check whether selected features are stable across folds or repeated samples.

• Use domain knowledge as well as statistical scores.

• For high-dimensional data, a fast filter can narrow the candidate set before a more expensive wrapper or model-based approach.

9. Short Exam Questions and Answers

1. Define feature selection.

Feature selection is the process of selecting a subset of relevant input features for model building while removing irrelevant or redundant features.

2. Name the three major feature-selection approaches.

Filter methods, wrapper methods, and embedded methods.

3. What is a filter method?

A method that ranks or selects features using statistical properties without repeatedly training a predictive model for each subset.

4. What is a wrapper method?

A method that evaluates candidate feature subsets using the performance of a trained model.

5. What is an embedded method?

A method that performs feature selection as part of model training, such as Lasso or tree-based selection.

6. Define a decision tree.

A supervised-learning model that predicts by applying a sequence of feature-based tests arranged as a tree.

7. What is entropy?

Entropy is a measure of uncertainty or impurity in the class labels of a node.

8. Define information gain.

Information gain is the reduction in entropy caused by splitting a dataset on a feature.

9. What is Gini impurity?

Gini impurity measures class mixing and is calculated as 1 minus the sum of squared class probabilities.

10. What is pruning?

Pruning removes or limits branches to reduce overfitting and improve performance on unseen data.

11. Differentiate classification and regression trees.

Classification trees predict categorical labels; regression trees predict numeric values.

12. What is data leakage in feature selection?

Data leakage occurs when information from validation or test data influences feature selection or model training, leading to overly optimistic evaluation.

10. Long-Answer Practice

• Explain filter, wrapper, and embedded feature-selection methods with examples, advantages, and limitations.

• Describe the construction of a decision tree and explain root nodes, internal nodes, branches, and leaves.

• Define entropy and information gain. Show how information gain is used to select a split.

• Compare ID3, C4.5, and CART.

• Explain overfitting in decision trees and discuss pre-pruning and post-pruning.

• Discuss the advantages and limitations of decision trees in data science.

Quick Revision

Term	Remember
Feature selection	Choose a subset of original inputs.

Term	Remember
Filter / Wrapper / Embedded	Statistical score / model-subset search / selection during training.
Entropy	Higher when class labels are more mixed; zero for a pure node.
Information gain	Parent entropy minus weighted child entropy; higher is preferred by ID3.
Gini	1 – sum of squared class probabilities; lower impurity is better.
Pruning	Controls tree complexity to reduce overfitting.